ZENELAIT LMS - COMPREHENSIVE PLATFORM MANUAL & FLOWCHART

Role-Based Capabilities & Package-Wise Feature Access Distribution

1. Package-Wise Feature Access Distribution (Basic vs. Standard vs. Premium)

Zenelait LMS functionalities are decoupled by subscription tiers. The platform owner manages these tiers at the tenant level. Features are locked or unlocked dynamically via API filters depending on the subscription level.

BASIC PACKAGE

- Student Workspace Dashboard
- Teacher Classroom Tools
- Basic Coursework Syllabus
- Daily Attendance Registry
- Simple HTML/PDF Grade sheets

Upgrade Subscription →

STANDARD PACKAGE

- All Basic Features Included
- Interactive Parent Audits
- Timetables & Exam Schedules
- Tuition Billing Engine
- Razorpay Payments & Invoices
- Email Notifications Alert Hub

Activate Premium →

PREMIUM PACKAGE

- All Standard Features Included
- Enterprise Dashboard Analytics
- Polymorphic AI Assistant Chat
- Gemini Contextual Pipeline API
- Verified Certificates PDF Engine
- Tab Blur Detection Quiz Proctor
- Granular Tenant Feature Toggles

PACKAGE MANAGEMENT RULES & SECURITY BOUNDARIES:

1. Endpoint Protection: Spring Security filter chains validate subscriber packages dynamically before executing service classes.
2. Multi-tenant Toggles: Ultra Super Admin toggles individual system features per academy tenant to bypass default packages.
3. Grade & Verification: Premium certificate engine compiles achievements using html2canvas & jsPDF with database-backed integrity hashes.
4. Wallet Integrations: Settle tuition fees securely via the Parent Wallet loaded online via Razorpay webhook transaction processors.

2. Platform System Administration Flow (Admins)

Shows the administration hierarchy from platform host level down to local school management. Admins register rosters, manage courses, configure timetables, and issue announcements.

ULTRA SUPER ADMIN

- Onboard new Organizations (Tenants)
- Enable/Disable Tenant Site logins
- Configure Subscription packages
- Override local Feature Flags

Provisions Tenant →

SUPER ADMIN (ACADEMY OWNER)

- Register local regular Admins
- Monitor Billing Subscriptions
- Track local Revenue Statistics
- Trigger invoice reminders

Registers Manager →

ADMIN (COORDINATOR)

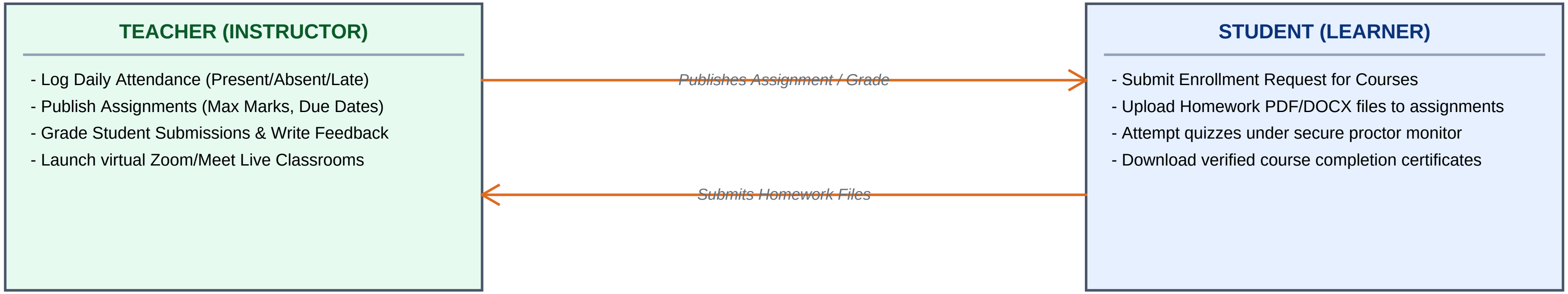
- Register Students/Teachers/Parents
- Create Courses & Departments
- Define Timetable Lecture slots
- Generate Fee Invoices (Single/Batch)

ADMINISTRATIVE WORKFLOW DETAILS & REST APIS:

```
- Create Organization tenant (Ultra Super Admin): POST /api/ultra-super-admin/organizations
- Enable/Disable Tenant Site login (Ultra Super Admin): PATCH /api/ultra-super-admin/organizations/{id}/toggle-active
- Register local administrator (Super Admin): POST /api/admin/users  (role=ADMIN)
- Configure Timetable slot (Admin): POST /api/admin/timetable  (Day, Start Time, End Time, Room, Course, Batch)
- Generate Fee Invoice (Admin): POST /api/admin/fees  (Single student)  ||  POST /api/admin/fees/batch-generate  (Batch-wide)
- Approve / Reject Course Enrollment Request (Admin): POST /api/admin/enrollments/requests/{id}/approve  ||  /reject
```

3. Academic Execution Flow (Teachers & Students)

Illustrates the classroom execution cycle. Teachers schedule lectures, register student presence, and grade deliverables. Students request enrollments, submit homework, and download verified certificates.

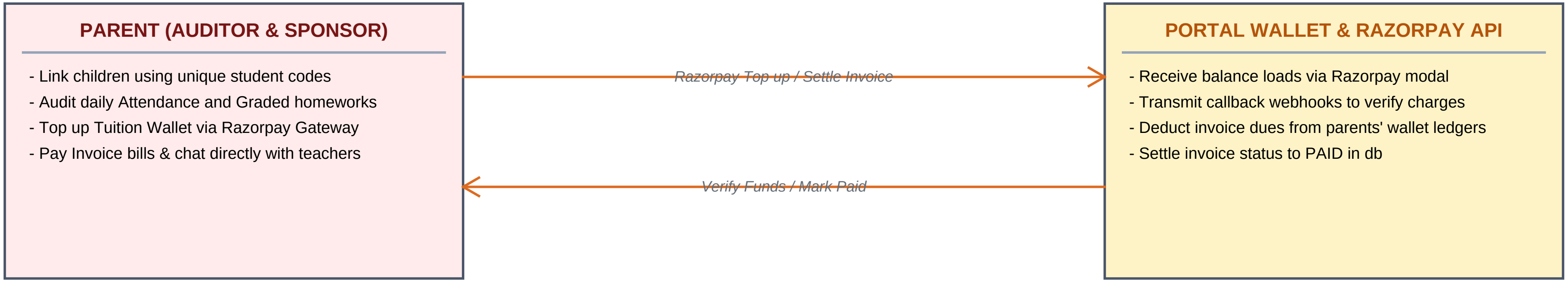


ACADEMIC LIFECYCLE & CORE REST APIS:

```
- Save Attendance Registry (Teacher): POST /api/teacher/attendance  (Logs status: PRESENT, ABSENT, LATE)
- Create Assignment (Teacher): POST /api/teacher/assignments  (Max marks, description, due date)
- Grade Student Submission (Teacher): PATCH /api/teacher/submissions/{id}/grade  (Input marks and text remarks)
- Start Timed Quiz (Student): POST /api/student/assessments/{id}/start
- Submit Quiz / Log Tab Blur Proctored violations (Student): POST /api/student/assessments/attempts/{id}/submit
- Download Verified Certificate PDF (Student): GET /api/student/certificates  (Generates secure SHA-256 integrity hash)
```

4. Parent Auditing, Message Threads & Wallet Billing Flow

Outlines the parent workflow. Parents link children using unique student codes, inspect their grades and attendance records, top up a custom portal wallet via Razorpay, settle tuition bills, and chat with teachers.



PARENT OPERATIONS & BILLING INTEGRATION REST APIS:

```
- Link Child Profile to parent user (Parent): POST /api/parent/children/link  (Params: studentCode)
- Fetch Linked Child Attendance history (Parent): GET /api/parent/children/{childId}/attendance
- Fetch Linked Child Grade Sheets (Parent): GET /api/parent/children/{childId}/grades
- Settle tuition bill from wallet balance (Parent): PATCH /api/parent/children/{childId}/fees/{id}/pay-from-wallet
- Start Text Message Thread with teacher (Parent): POST /api/parent/messages/conversations  (Params: teacherId)
- Send chat message payload (Parent/Teacher): POST /api/parent/messages/send
```

5. AI Assistant Context Gathering Pipeline & System Sequence

Illustrates the Google Gemini Context Pipeline. When a user queries the AI chat, the system pulls custom database views based on the user's active role, injecting it as structured prompt context.

Step 1: User Queries chatbot --> POST /api/ai/chat. Server receives message payload and validates active JWT token.

Step 2: Fetch Role Context --> Student: Gathers grades, attendance. Parent: Gathers linked children stats. Teacher: Gathers courses.

Step 3: Compile System Prompt --> Injects DB records as JSON context. Appends directive instruction e.g. 'Review this child's attendance'.

Step 4: Forward to Gemini API --> Sends context + user message to Google Gemini API (gemini-2.5-flash) using application.properties key.

Step 5: Dispatch Answer to UI --> Receives LLM completion response and streams it back to user dashboard chat client.

TRANSACTION SEQUENCE SUMMARY:

1. Tenant Creation: Ultra Super Admin -> registers organization -> enables tenant domain -> setups database configuration.
2. Billing Activation: Super Admin -> selects package (Basic/Standard/Premium) -> completes payment -> enables local APIs.
3. Roster Management: Admin -> registers teachers/students/parents -> configures departments, batches, and course structures.
4. Operation Loop: Teachers register attendance & publish tasks -> Students submit files -> Parents inspect progress & pay bills.
5. Graduation/Archive: Admin promotes batch students -> exports transcripts -> issues verified SHA-hash certificates.